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Stablecoin–FX Linkages:
Evidence from Fiat–Stablecoin Pair Listings on a Global Exchange

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As the issuance and use of USD stablecoins have expanded, policy interest in the linkages between USD stablecoin markets and conventional foreign exchange markets has increased. In particular, trading USD stablecoins against non-U.S. fiat currencies is similar in nature to trading dollar-denominated assets with those currencies and, depending on market structure, may translate into actual foreign exchange transactions and exchange rate movements. Existing studies typically assume a market structure in which global intermediaries with access to both markets participate as counterparties in fiat–USD stablecoin transactions. Under such a structure, global intermediaries can sell USD stablecoins and subsequently adjust the non-U.S. fiat currency positions they receive in the FX market. In practice, however, fiat–USD stablecoin transactions do not always take place under such a market structure. This paper focuses on this variation and examines how the participation of global intermediaries affects linkages between the two markets.

To identify this effect, we exploit the introduction of specific fiat–USD stablecoin trading pairs on Binance, a global exchange whose liquidity providers are already active in both stablecoin and FX markets, as an event that creates a market structure in which these global intermediaries can participate directly as counterparties. Following the introduction of a trading pair, global intermediaries can draw on global stablecoin liquidity to meet the demand of investors holding the relevant fiat currency and adjust the resulting fiat currency positions in the FX market. This market structure can strengthen both price integration, whereby stablecoin prices converge more closely to spot exchange rates, and shock transmission, whereby demand shocks originating in the stablecoin market are transmitted to exchange rates through FX transactions.

The empirical results support both channels. In terms of price integration, USD stablecoin premia decline significantly following the introduction of Binance trading pairs, and stablecoins flow from Binance into local markets when local premia exceed those on Binance. In terms of shock transmission, stablecoin premia have only a limited effect on the relevant U.S. dollar exchange rate before the introduction of a trading pair, whereas after pairing, higher premia are associated with significant local currency depreciation. Moreover, net buyer-initiated order flow, which captures demand pressure in the stablecoin market, is significantly associated with depreciation of the paired currencies. By contrast, in Korea, where no such Binance trading pair is available, net buying pressure is primarily reflected in higher stablecoin premia and has no significant effect on the exchange rate.

These findings suggest that linkages between USD stablecoin and FX markets could strengthen as Korea's digital asset market structure evolves, particularly with greater participation by corporations and foreign investors. Accordingly, digital asset regulation should be considered together with efforts to promote the internationalization of the Korean won and deepen FX market liquidity to strengthen the market's capacity to absorb shocks.

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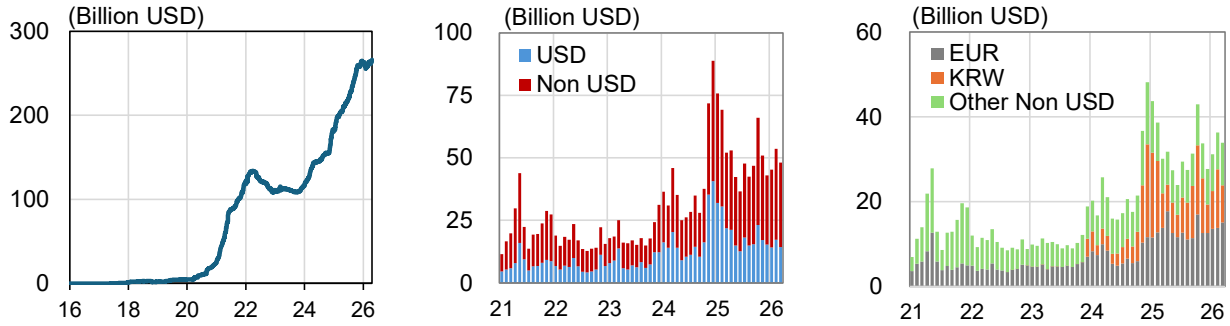
■ We would like to express our sincere appreciation to Kyoungsoo Yoon, Jae Hyun Yoo, Yong O Kwon for their comments and suggestions. Any errors in this paper are solely the responsibility of the authors.

Bank of Korea

Ⅰ. Introduction

1. As the issuance and use of USD stablecoins have expanded, policymakers have shown growing interest in their links with traditional foreign exchange markets. USD stablecoin issuance has risen steadily since 2020 [Figure 1(a)], while trading against non-U.S. currencies, including the Korean won, euro, and Brazilian real, has also increased[Figure 1(b)(c)]. Because USD stablecoins are pegged to the U.S. dollar, purchasing them with a non-U.S. currency is economically similar to converting that currency into a dollar-denominated asset. Understanding the extent to which stablecoin trading leads to actual FX transactions and the linkage between stablecoin and FX markets has therefore become an important policy issue.

[Figure 1] USD Stablecoin Issuance and Trading Volume



Notes: USD stablecoins refer to USDT and USDC. Issuance is measured as the sum of their market capitalizations reported by CoinMarketCap. Trading volume is calculated using CryptoCompare API data by aggregating daily fiat-denominated USDT and USDC trading volume across 60 global cryptocurrency exchanges, separately for USD, non-USD currencies, and individual fiat currencies.

Data: CoinMarketCap, CryptoCompare

2. We define the linkage between USD stablecoin and FX markets as the extent to which trading USD stablecoins against non-U.S. fiat currencies (hereafter, fiat–stablecoin trading) is connected to actual FX transactions. This linkage has two main dimensions: price integration and shock transmission. Price integration refers to the narrowing of deviations between the local-currency price of a USD stablecoin and the corresponding spot exchange rate through arbitrage across the two markets. Shock transmission refers to the extent to which demand for USD stablecoins in local currency induces FX transactions and affects exchange rates.

3. Existing studies show that USD stablecoin and FX markets can become linked when global intermediaries[1] with access to both markets act as counterparties in fiat–stablecoin trades (Aldasoro et al., 2026). When stablecoin demand creates a gap between local stablecoin prices and spot exchange rates, arbitrage incentives increase, and global intermediaries narrow the gap through cross-market trading (price integration). In doing so, they trade in FX markets to rebalance the dollar and local-currency positions generated by their arbitrage activity, transmitting shocks from

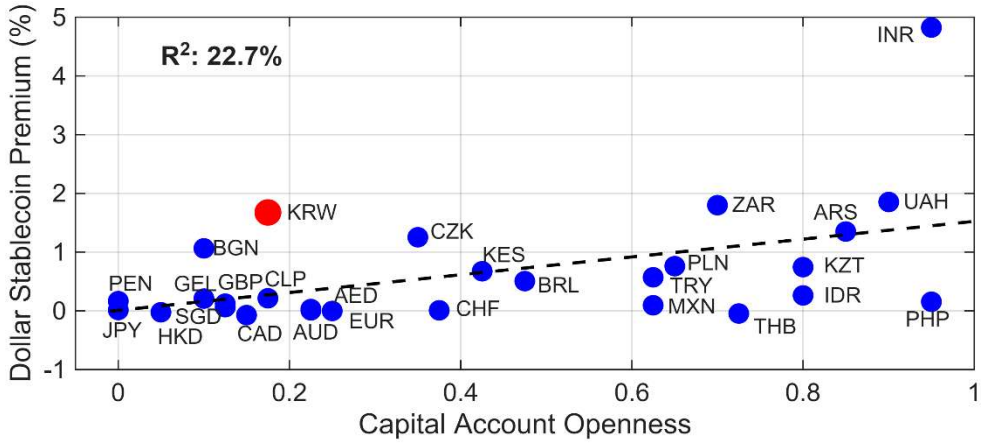
[1] In this paper, we refer to professional financial institutions that operate in both global USD stablecoin and FX markets, providing liquidity and conducting arbitrage across the two markets, as global intermediaries or professional market makers.

stablecoin markets to FX markets (shock transmission).[2]

4. Existing studies assume a market structure in which global intermediaries act as counterparties in fiat–stablecoin trades. In practice, however, not all trades take place under this structure. When USD stablecoins are traded only among investors within the same currency area, demand does not reach global intermediaries, and the arbitrage and position-adjustment mechanisms described in prior studies do not operate. Thus, even for otherwise similar fiat–stablecoin trades, the linkage between stablecoin and FX markets may depend on the underlying market structure. The effects of this market structure on price integration and shock transmission, however, have not been systematically examined.

5. The importance of market structure is illustrated by cross-country patterns in stablecoin premia, which measure the deviation between USD stablecoin prices and spot exchange rates. Previous studies argue that lower capital-account openness constrains arbitrage and leads to higher stablecoin premia. Consistent with this view, countries with more restrictive capital accounts tend to exhibit larger premia [Figure 2].

[Figure 2] USD Stablecoin Premia and Capital Account Openness across Currencies



Notes: The USD stablecoin premium is calculated as $(P_{LC}^{USDT} - S_{LC}/S_{LC})$, where (P_{LC}^{USDT}) is the price of USDT in the relevant local currency and (S_{LC}) is the corresponding exchange rate against the U.S. dollar. The figure reports the median premium since 2022. Capital account openness is measured using the latest available observation for each country, with Germany used as the proxy for the euro area.

Data: CryptoCompare, Bergant et al. (2026)

6. However, Korea stands out as a notable exception, exhibiting a sizable USD stablecoin premium of 1.67% despite its relatively open capital account. This is comparable to the premia observed in Ukraine (1.86%) and South Africa (1.80%), where capital controls are more stringent. Korea

[2] When the local-currency price of a USD stablecoin exceeds the spot exchange rate, an investor can earn an arbitrage profit by converting local currency into U.S. dollars, using those dollars to purchase the stablecoin, and then selling it for local currency. This arbitrage requires access to both the FX market and the global USD stablecoin market. Limited access to the FX market raises the cost of converting local currency into dollars, while limited access to the global stablecoin market raises the cost of converting those dollars into stablecoins.

therefore deviates from the general relationship between capital controls and stablecoin premia.[3] This suggests that stablecoin premia may depend not only on capital account openness but also on the market structure through which stablecoins are traded.[4]

7. We use the introduction of direct USD stablecoin trading against a given fiat currency on Binance (hereafter, Binance fiat pairing) as an event that changes market structure. This setting allows us to examine how the structure of fiat–stablecoin trading affects the linkage between stablecoin and FX markets.[5] Once Binance introduces a fiat–stablecoin pair, global intermediaries can participate as counterparties in trades involving that currency.[6] We analyze how this change affects price integration and shock transmission, and discuss its implications for FX markets and policy.

[3] In a cross-country regression of capital account openness on USD stablecoin premia, the R^2 rises from 22.7% when Korea is included to 27.4% when it is excluded, confirming that Korea is an outlier in this relationship.

[4] Korea's cryptoasset market is dominated by retail investors because participation by corporations and foreign investors is limited. As a result, arbitrage by global intermediaries with access to both USD stablecoin and FX markets is constrained, weakening the market-adjustment mechanism that would otherwise narrow price discrepancies. Consequently, deviations between the KRW price of USD stablecoins and the spot exchange rate may be large in absolute terms or persist for extended periods. This applies to both premia and discounts.

[5] Beyond their links with FX markets, USD stablecoins may also affect traditional financial markets through several channels. In bond markets, stablecoin issuance can increase demand for reserve assets such as government securities and money market funds, with potential effects on yields and market liquidity (Barthélémy et al., 2023; Ahmed and Aldasoro, 2025; Kim, 2025). In FX markets, global intermediaries may affect exchange rates when they raise U.S. dollar funding, including through FX swaps, in response to stablecoin demand shocks (Aldasoro et al., 2026). Using Korean data, Park (2025) also documents a significant relationship between USD stablecoin premia and the USD/KRW exchange rate during 2024–2025. Our study uses the introduction of fiat–stablecoin trading on Binance to provide evidence on a previously unexplored channel: how changes in market structure shape linkages across markets.

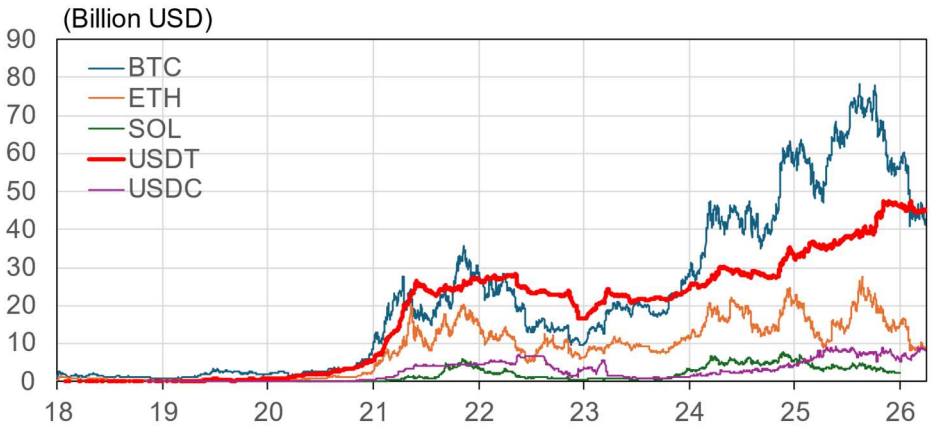
[6] This mechanism is consistent with Evans and Lyons (2002), Gabaix and Maggiori (2015), and Itskhoki and Mukhin (2021), who show that currency positions accumulated by intermediaries while serving customer trades can affect FX markets. More specifically, we empirically examine whether the stablecoin–FX linkage through global intermediary arbitrage proposed by Aldasoro et al. (2026) operates through the market structure created by Binance's introduction of fiat–stablecoin trading.

Ⅱ. Mechanisms Linking Binance Fiat Pairing to FX Markets

Ⅱ-1. Market Structure Changes from Binance Fiat–Stablecoin Pair Listings

8. Binance is a central hub of global USD stablecoin liquidity. It is the world's largest centralized cryptocurrency exchange by spot trading volume, accounting for about 39% of global centralized exchange volume in 2025. Binance also holds about 69% of the USDT and USDC balances deposited at major centralized exchanges, making it the main venue for global USD stablecoin liquidity [Figure 3], [Figure 4].

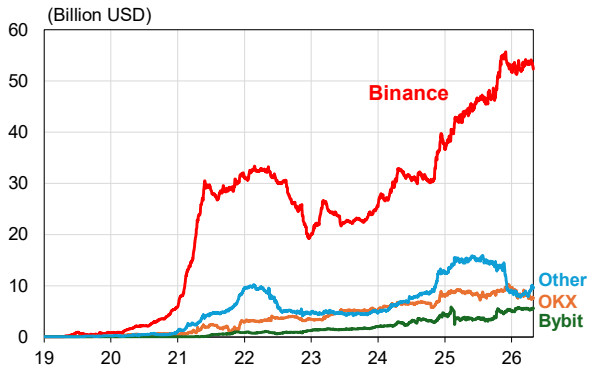
[Figure 3] Binance's Major Cryptoasset Holdings



Notes: For wallets classified as Binance by Chainalysis, balances are estimated by cumulatively summing net inflows of Bitcoin (BTC), Ethereum (ETH), Solana (SOL), USDT, and USDC over time and valuing them in U.S. dollars using contemporaneous market prices.
Data: Chainalysis, CoinMarketCap, Authors' calculations

[Figure 4] USD Stablecoin Balances at Major Crypto Exchanges and Their Shares

(a) Balance over Time by Exchanges



(b) Balances and Shares at End-2025

Exchange	Balance (USD bn)	Share (%)
Binance	51.83	(69.11)
OKX	9.58	(12.77)
Bybit	4.96	(6.62)
Upbit	0.23	(0.30)
Bithumb	0.09	(0.12)
Other	8.31	(11.08)

Notes: Balances are estimated by classifying wallets identified by Chainalysis as centralized exchanges into Binance, OKX, Bybit, and other exchanges, and cumulatively summing net USDT and USDC inflows for each group over time. Each exchange's share is calculated relative to the total USDT and USDC balance held in wallets classified as centralized exchanges.
Data: Chainalysis, CoinMarketCap, Authors' calculations

9. Building on this liquidity, Binance supports direct trading between a wide range of fiat currencies and USD stablecoins. Once trading against a particular fiat currency becomes available, investors holding that currency can buy and sell USD stablecoins directly on Binance without first converting it into another currency or routing the transaction through another cryptoasset. To facilitate such trading, Binance supports deposits, withdrawals, and payments in the relevant fiat currency, thereby directly connecting it to the global USD stablecoin market[7].

10. Binance also supports trading between many fiat currencies and cryptoassets other than USD stablecoins. However, fiat–stablecoin trades account for a large share of its total fiat-denominated cryptoasset trading[Table 1]. For the Ukrainian hryvnia and South African rand, only USD stablecoin pairs are available, so their stablecoin shares are 100%. USD stablecoins also account for more than half of fiat-denominated trading in the Argentine peso (92.5%), Brazilian real (63.3%), and Polish zloty (74.4%), despite the availability of fiat pairs for other cryptoassets such as Bitcoin and Ethereum. This is because USD stablecoins serve as quote currencies for trading other cryptoassets on Binance.

[Table 1] Binance Trading Volume by Major Fiat Currency in 2025

(USD bn, %)

	Argentina	Brazil	Euro Area	Japan	Poland	Türkiye	Ukraine	U.S.	South Africa
USDT	0.7	14.1	9.7		0.2	23.4	0.0	0.1	0.2
USDC		0.3	9.4		0.5	2.5		0.0	
BTC	0.1	5.1	13.2	5.0	0.2	4.2		0.0	
ETH	0.0	1.8	10.6	3.6	0.1	5.0			
SOL		0.6	3.7	0.5		1.7			
XRP		0.3	4.0	12.1		5.6			
Others		0.6	6.1	4.4		102.0			
Sum	0.7	22.8	56.7	25.6	1.1	144.2	0.0	0.1	0.2
(SC %)	(92.5)	(63.3)	(33.6)	(0.0)	(74.4)	(18.0)	(100.0)	(96.4)	(100.0)

Notes: This table reports the USD-equivalent trading volume of fiat-denominated cryptoasset pairs traded on Binance in 2025. USDT and USDC denote USD stablecoins, while BTC, ETH, SOL, and XRP denote Bitcoin, Ethereum, Solana, and Ripple, respectively. Others includes all other cryptoassets. Figures in parentheses indicate the share of total trading volume accounted for by USD stablecoins (USDT and USDC).

Data: Binance, Bloomberg, Authors' calculations

11. Binance is both a major venue for global USD stablecoin liquidity and a market where USD stablecoins are widely traded against fiat currencies. When Binance introduces trading against a specific fiat currency, it does more than expand access. It changes the market structure through which stablecoins are traded in that currency.

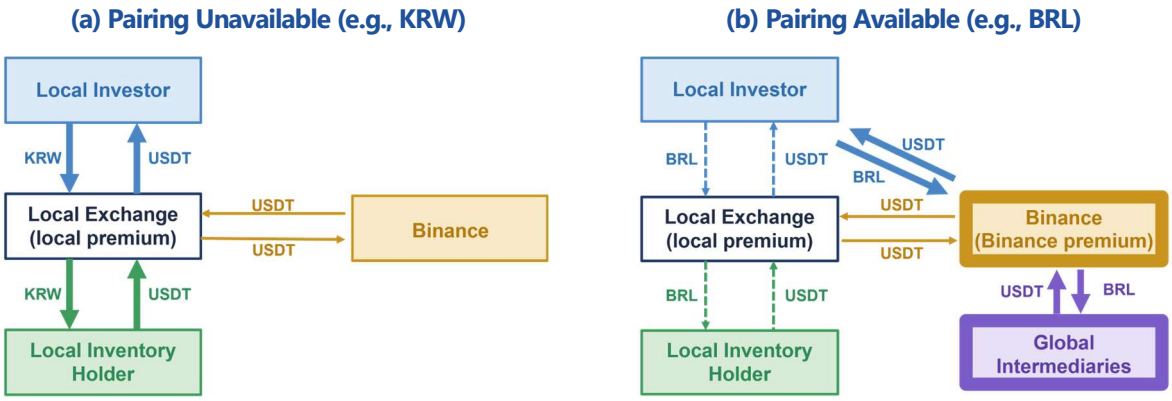
[7] Binance users can deposit fiat currency into their accounts through bank transfers or Visa cards and use these funds to trade cryptoassets on the platform (Binance, 2024).

12. [Figures 5(a) and 5(b)] illustrate the difference in market structure between a currency without Binance fiat pairing, the Korean won (KRW), and a currency with pairing, the Brazilian real (BRL).

13. When fiat–stablecoin pairing is unavailable, demand for USD stablecoins in that currency is cleared among investors within the same currency area. Because investors cannot purchase USD stablecoins directly with the currency on Binance, demand on local exchanges is met by stablecoins supplied by investors in the same currency area. The exchange between the fiat currency and USD stablecoins therefore takes place within the local market.

14. When direct trading is available on Binance, demand for USD stablecoins in that currency is transmitted directly to professional market makers[the Global Intermediaries in Figure 5(b)]**.** Investors can deposit the fiat currency on Binance and use it to purchase USD stablecoins, while professional market makers act as counterparties, receive the fiat currency, and supply the corresponding amount of stablecoins. The counterparties to fiat–stablecoin trades are therefore no longer limited to investors within the same currency area but also include professional market makers. As a result, local stablecoin demand is transmitted directly to the global market rather than being absorbed within the local market.

[Figure 5] Changes in Trading Structure Following Binance Fiat–Stablecoin Pairing



Ⅱ-2. Mechanism Linking Binance Fiat Pairing to Stronger FX Market Linkages

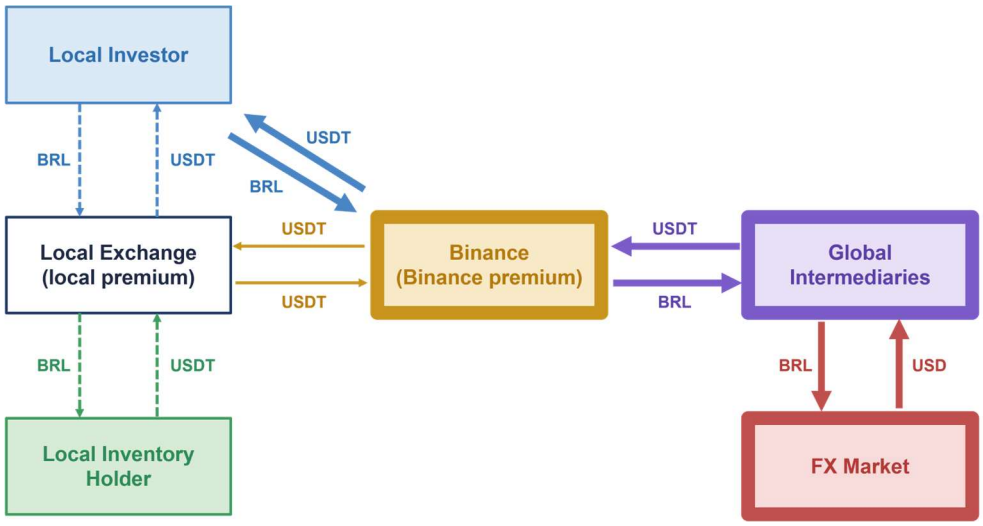
15. Once Binance introduces USD stablecoin trading against a specific fiat currency, the market shifts from one centered on investors within the same currency area to one involving global intermediaries. This change strengthens the link between fiat–stablecoin trades and actual FX transactions through two channels: price integration and shock transmission.

16. Binance's fiat–stablecoin markets rely on professional market makers participating in its *Fiat Liquidity Provider Program*. These market participants meet minimum trading and liquidity provision requirements and act as counterparties in fiat–stablecoin trades.[8] When stablecoin prices deviate from spot exchange rates, they have incentives to arbitrage across the two markets and trade

[8] Binance encourages liquidity provision through its *Fiat Liquidity Provider Program* by offering maker fee rebates to participants that meet minimum trading volume and liquidity provision requirements (Binance, 2021; 2023).

in the FX market to manage the resulting currency positions. These activities help align stablecoin prices with spot exchange rates and support market liquidity.

[Figure 6] Stablecoin–FX Market Linkages under Fiat–Stablecoin Pairing



17. This structure strengthens shock transmission. Professional market makers have incentives to trade in the FX market to manage the currency positions generated by fiat–stablecoin trades and to reduce price gaps across the two markets. Demand shocks in fiat–stablecoin markets can therefore be transmitted to the FX market through market makers' position management. [Figure 6] illustrates this mechanism. After Binance fiat pairing, professional market makers absorb local investors' demand for USD stablecoins and generate FX transactions as they manage the resulting positions. [Figure 7] shows the corresponding balance sheet adjustment. When an investor purchases USDT with Brazilian real (BRL), the market maker supplies USDT and receives BRL. Its balance sheet therefore shifts from matched dollar assets and liabilities to BRL assets and dollar liabilities, creating a currency mismatch[Figure 7(a)]→[Figure 7(b)]. To close this mismatch, the market maker has an incentive to sell BRL and buy U.S. dollars in the FX market[Figure 7(b)]→[Figure 7(c)]. These transactions can affect the exchange rate.

[Figure 7] Changes in Market Makers' Balance Sheets from Fiat–Stablecoin Trading

(a)

Asset	Liability
USDT	USD

(b)

Asset	Liability
BRL (USDT→BRL)	USD

(c)

Asset	Liability
USDT (BRL→USD) (USD→USDT)	USD

18. This change in market structure also strengthens price integration. Before Binance fiat pairing, demand for USD stablecoins in a given currency is met mainly by the existing holdings of investors within the same currency area[Figure 5(a)], making supply on local exchanges relatively inelastic. Investors may also convert the local currency into another fiat currency supported for stablecoin trading and purchase USD stablecoins on another exchange. However, this route involves conversion costs, transfer costs, and settlement delays, and is therefore less elastic than supply provided by professional market makers with direct access to global USD stablecoin liquidity. After pairing, local demand can also be met through global intermediaries[Figure 5(b)]. Professional market makers can source and immediately supply USD stablecoins from global liquidity, temporarily absorb the resulting fiat currency positions on their balance sheets, and manage them in the FX market. This allows them to respond more elastically to increases in local demand. As a result, local stablecoin prices converge more quickly to spot exchange rates, limiting increases in stablecoin premia.

19. In sum, Binance fiat pairing strengthens both price integration and shock transmission. For price integration, the participation of global intermediaries makes USD stablecoin supply on local exchanges more elastic, limiting increases in stablecoin premia. For shock transmission, market makers manage the currency positions generated by fiat–stablecoin trades in the FX market, allowing demand shocks in the USD stablecoin market to be transmitted to the FX market.

Ⅱ-3. Motivating Evidence of the Mechanism

20. The mechanism implies different empirical predictions for price integration and shock transmission when Binance introduces USD stablecoin trading against a specific fiat currency. For price integration, the response of local stablecoin premia to cryptoasset demand shocks should weaken after pairing. For shock transmission, the effect of the same shocks on spot exchange rates should increase as professional market makers trade in the FX market to manage their positions.

21. To examine these predictions, we compare how cryptoasset demand shocks affect USD stablecoin premia and exchange rates in Korea, where Binance fiat pairing is unavailable, and Brazil, where it is available. Specifically, we estimate the effects of Bitcoin search intensity from Google Trends on local stablecoin premia and exchange rate changes in the two countries. Bitcoin search intensity is used as a proxy for cryptoasset investment demand because higher search activity may increase demand for USD stablecoins used in cryptoasset trading, while having little direct link to dollar demand in the FX market. In Korea, a one-standard-deviation increase in Bitcoin search intensity raises the USD stablecoin premium by about 0.85 percentage points, with no statistically significant effect on the USD/KRW exchange rate. In Brazil, the same shock raises the premium by about 0.11 percentage points, a smaller response than in Korea, but increases the USD/BRL exchange rate by about 0.12%, indicating a significant depreciation of the real. Bitcoin search intensity also has greater explanatory power for stablecoin premia in Korea and for exchange rate changes in Brazil.

[Table 2] Transmission of Bitcoin Search Demand to Stablecoin and FX Markets: Korea vs. Brazil

	USD Stablecoin Premia (%)		Exchange Rate Change (%)	
	Korea	Brazil	Korea	Brazil
	(1)	(2)	(3)	(4)
$GT^{Bitcoin}$	0.850***	0.109***	0.039	0.118**
	(0.162)	(0.017)	(0.037)	(0.060)
Partial R^2(%)	14.72	6.53	0.33	0.74
Observations	334	510	334	510

Notes: Each column reports regression results for Korea and Brazil using weekly data. The dependent variables in columns (1)–(2) are the volume-weighted average USD stablecoin premia (%) in Korea and Brazil, respectively. The dependent variables in columns (3)–(4) are the log changes in the nominal exchange rates against the U.S. dollar (%), where positive values indicate local currency depreciation. $GT^{bitcoin}$ is the standardized Google Trends index for Bitcoin searches in each country. Partial R^2 is the additional share of variation explained by $GT^{bitcoin}$ after controlling for global factors, including the U.S. dollar index and the VIX, and stablecoin fixed effects. Robust standard errors are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

22. These results are consistent with the proposed mechanism. Binance fiat pairing links local stablecoin demand to global intermediaries, dampening stablecoin premia on local exchanges through price integration while transmitting demand shocks to the FX market through market makers' position adjustments. The next section describes the data and construction of the main variables used to test these hypotheses, and then examines how fiat–stablecoin pairing affects the linkage between the two markets.

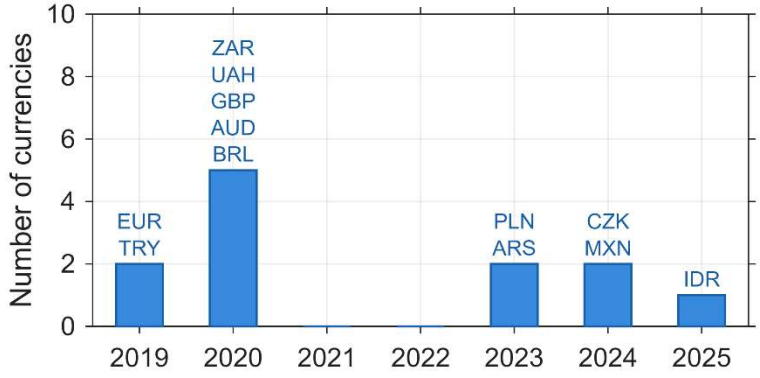
Ⅲ. Data and Key Variables

23. The key variables used in this study are ① the start date of Binance fiat–stablecoin pairing, ② USD stablecoin flows across exchanges, ③ net buying pressure in USD stablecoins, and ④ the USD stablecoin premium, which measures the deviation between USD stablecoin prices and spot exchange rates. This section describes how these variables are constructed and presents related stylized facts, including summary statistics.

Ⅲ-1. Start Dates of Binance Fiat–Stablecoin Pairing

24. Using Binance transaction data, we define the start date of fiat–stablecoin pairing for a given fiat currency as the first date on which price and trading volume data are observed for direct trading between that currency and a USD stablecoin, either USDT or USDC, on Binance. During the sample period from 2017 to 2026, the Turkish lira and the euro were the first currencies to be paired, followed by the South African rand, British pound, Australian dollar, Brazilian real, and other fiat currencies. As of end-April 2026, a total of 16 fiat currencies had been directly traded against USD stablecoins on Binance. Because our analysis requires fiat–stablecoin trading data from multiple exchanges, including Binance, 12 currencies satisfy the sample selection criteria.[9] Pairing start dates are distributed across several years from 2019 to 2025[Figure 8], allowing us to exploit cross-country variation in the timing of pairing.

[Figure 8] Start Year of Binance Fiat–Stablecoin Pairing by Currency



Note: The start date of direct pairing is defined as the first date on which price and trading volume data are recorded for the relevant fiat currency–USD stablecoin (USDT) pair on Binance. The figure aggregates these start dates by year.
Data: CryptoCompare

[9] A total of 16 fiat currencies were paired on Binance, in the following order: the Turkish lira, euro, South African rand, Ukrainian hryvnia, British pound, Australian dollar, Nigerian naira, Brazilian real, Polish zloty, Romanian leu, Argentine peso, Mexican peso, Czech koruna, Colombian peso, U.S. dollar, and Indonesian rupiah. Among these, the Colombian peso, Nigerian naira, and Romanian leu are excluded because price data from cryptoasset exchanges other than Binance are highly noisy. The U.S. dollar is also excluded because it serves as the reference currency for USD stablecoins.

Ⅲ-2. USD Stablecoin Flows

25. Using daily blockchain data on transfers of USD stablecoins (USDT and USDC) across exchanges, we examine whether price integration occurs through USD stablecoin flows from Binance. Specifically, when Binance supports direct trading between a given fiat currency and USD stablecoins, we test whether the premium differential between a local exchange and Binance is associated with net inflows of USD stablecoins from Binance into the corresponding fiat–stablecoin market. This provides direct evidence that price integration occurs through actual movements of USD stablecoins.

26. To construct this measure, we allocate USD stablecoin transfers between Binance and other exchanges to the fiat–stablecoin markets of those exchanges and calculate outflows from Binance to each market. For an exchange that supports only one fiat currency, all inflows from and outflows to Binance are assigned to that fiat–stablecoin market. For an exchange that supports multiple fiat currencies, the volume of USD stablecoins transferred between Binance and the exchange is allocated across currencies according to each currency's share of the exchange's fiat–stablecoin trading volume. This allocation assumes that these trading shares reflect USD stablecoin demand across currency markets on the exchange. We then use the resulting inflows and outflows to calculate net inflows from Binance into each fiat–stablecoin market.

27. Net inflows from Binance into fiat–stablecoin market i are defined as USD stablecoin inflows from Binance into the market minus USD stablecoin outflows from the market to Binance. Inflows from Binance into market i are calculated by allocating USD stablecoins transferred from Binance to each destination exchange according to the destination exchange's fiat currency shares in USD stablecoin trading volume and then summing across all destination exchanges, as follows.

$$Inflow_{ijt}^{Binance} = \sum_{kd} Flow_{jt}^{Binance \to kd} \omega_{ikd}$$

Outflows from fiat–stablecoin market i to Binance are calculated by allocating the volume of USD stablecoins transferred from each origin exchange to Binance according to the origin exchange's fiat currency shares in USD stablecoin trading volume and then summing across all origin exchanges, as follows.

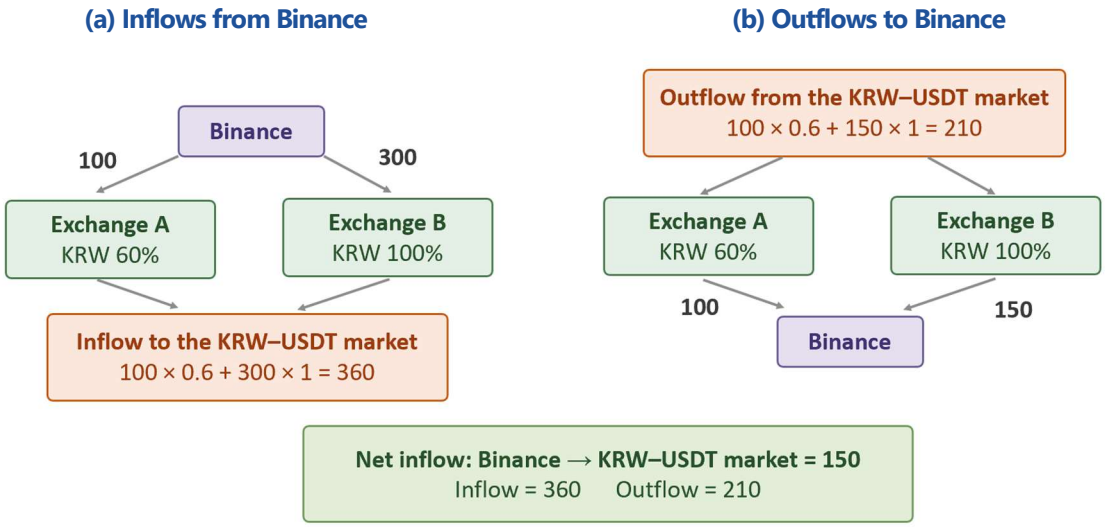
$$Outflow_{ijt}^{Binance} = \sum_{ko} Flow_{jt}^{ko \to Binance} \omega_{iko}$$

Net inflows from Binance into fiat–stablecoin market i are defined as the difference between inflows and outflows, as follows.

$$\begin{aligned} Net\ Inflow_{ijt}^{Binance} &= Inflow_{ijt}^{Binance} - Outflow_{ijt}^{Binance} \\ &= \sum_{kd} Flow_{jt}^{Binance \to kd} \omega_{ikd} - \sum_{ko} Flow_{jt}^{ko \to Binance} \omega_{iko} \end{aligned}$$

28. [Figure 9] illustrates how net inflows from Binance into a given fiat–stablecoin market are constructed. USD stablecoins transferred from Binance to each exchange are allocated to inflows into the corresponding fiat–stablecoin markets based on the destination exchange's trading shares by fiat currency. Conversely, USD stablecoins transferred from each exchange to Binance are allocated to outflows from the corresponding markets based on the origin exchange's trading shares. Net inflows from Binance are then calculated as inflows minus outflows.

[Figure 9] Net Inflows from Binance into Fiat–Stablecoin Markets



Note: The arrows indicate the direction of USD stablecoin transfers, and the numbers shown on the arrows indicate the transfer amounts.

Ⅲ-3. USD Stablecoin Net Buying Pressure

29. We use net buying pressure in USD stablecoins on Binance as a proxy for the fiat currency positions borne by market makers as counterparties to fiat–stablecoin trades. Net buying pressure for trades between fiat currency i and USD stablecoin j is defined as the difference between buyer-initiated and seller-initiated trading volume, scaled by the previous period's trading volume.

$$Net\ Buying\ Pressure_{ijt} = \frac{Buyer\ Initiated\ Trade_{ijt} - Seller\ Initiated\ Trade_{ijt}}{Trading\ Volume_{ij,t-1}}$$

When buyer-initiated purchases of USD stablecoins using a given fiat currency exceed seller-initiated sales on Binance, market makers on the other side of the trades temporarily hold a long position in that fiat currency equal to the imbalance. Under the mechanism described above, market makers have an incentive to manage these currency positions in the FX market. Net buyer-initiated order flow can therefore be interpreted as an indirect measure of pressure on market makers to sell the fiat currency and buy U.S. dollars in the FX market. Binance identifies whether each trade is buyer- or seller-initiated, allowing us to construct net buyer-initiated order flow by fiat currency. Over the sample period, net buyer-initiated order flow averaged 4.9% of the previous day's trading volume,

suggesting that market makers held fiat currency positions equal to about 4.9% of the previous day's trading volume on average.[10]

30. As a counterfactual, we also construct a measure of net buying pressure in KRW–USD stablecoin trading on Korean exchanges, where fiat–stablecoin pairing is unavailable on Binance. The economic interpretation of this variable differs from that of net order flow on Binance. Because fiat–stablecoin trades on Korean exchanges are not intermediated by global intermediaries, net buyer-initiated order flow does not represent KRW positions held by market makers. Instead, net order flow on Korean exchanges primarily measures KRW-based demand pressure for USD stablecoins.

31. Comparing net order flow on Binance with that on Korean exchanges allows us to distinguish between otherwise similar USD stablecoin demand shocks that are transmitted through global intermediaries and those that are not.

Ⅲ-4. USD Stablecoin Premia

32. We use the USD stablecoin premium to measure the degree of price integration between USD stablecoin and FX markets. The premium captures the relative difference between the cost of obtaining USD stablecoins using a given fiat currency and the cost of obtaining U.S. dollars in the FX market, and is defined as follows.

$$\underbrace{USD\ Stablecoin\ Premia}_{parity\ deviation} \equiv \frac{P_{ijkt}^{SC} - S_{it}}{S_{it}} \times 100$$

P_{ijkt}^{SC} denotes the price of USD stablecoin j, denominated in local currency i, on exchange k at time t, and S_{it} denotes the exchange rate of currency i against the U.S. dollar. We calculate the premium using the daily volume-weighted average stablecoin price by currency, constructed from exchange-level daily closing prices and trading volumes obtained from CryptoCompare and other sources, together with exchange rates against the U.S. dollar from Bloomberg.[11] A positive USD stablecoin premium indicates that obtaining USD stablecoins using the local currency is more costly than obtaining U.S. dollars in the spot FX market.

[10] Average net buyer-initiated order flow varies across currencies, ranging from 1.8% for the Ukrainian hryvnia to 16.1% for the Indonesian rupiah.

[11] For the Argentine peso and Ukrainian hryvnia, official spot exchange rates do not adequately reflect market exchange rates because of FX controls and country-specific factors. Following Graf von Luckner, Reinhart, and Rogoff (2023), we therefore use parallel exchange rates as proxies for spot exchange rates for these currencies.

33. [Table 3] reports the median USD stablecoin premium for each currency. Positive premia are observed for most currencies, with a median of about 0.8% across the 30 currencies. The premium varies considerably across currencies. It is close to zero for the Canadian dollar, euro, and Swiss franc, but exceeds 4% for some currencies, including the Indian rupee and Zambian kwacha.

[Table 3] Median USD Stablecoin Premium by Currency

Currency	Premium (%)	Currency	Premium (%)	Currency	Premium (%)
Canada	-0.07	Singapore	0.12	Kazakhstan	0.75
Thailand	-0.05	Philippines	0.15	Poland	0.76
Hong Kong	-0.02	Peru	0.16	Bulgaria	1.07
Euro Area	0.00	Georgia	0.21	Czech Rep.	1.25
Switzerland	0.01	Chile	0.21	Argentina	1.35
UAE	0.01	Indonesia	0.26	**Korea**	**1.67**
Japan	0.01	Taiwan	0.49	South Africa	1.80
Australia	0.03	Brazil	0.51	Ukraine	1.86
U.K.	0.06	Türkiye	0.57	Zambia	4.20
Mexico	0.10	Kenya	0.67	India	4.82

Note: The estimates are based on USDT and cover only the period since 2022.
Data: CryptoCompare

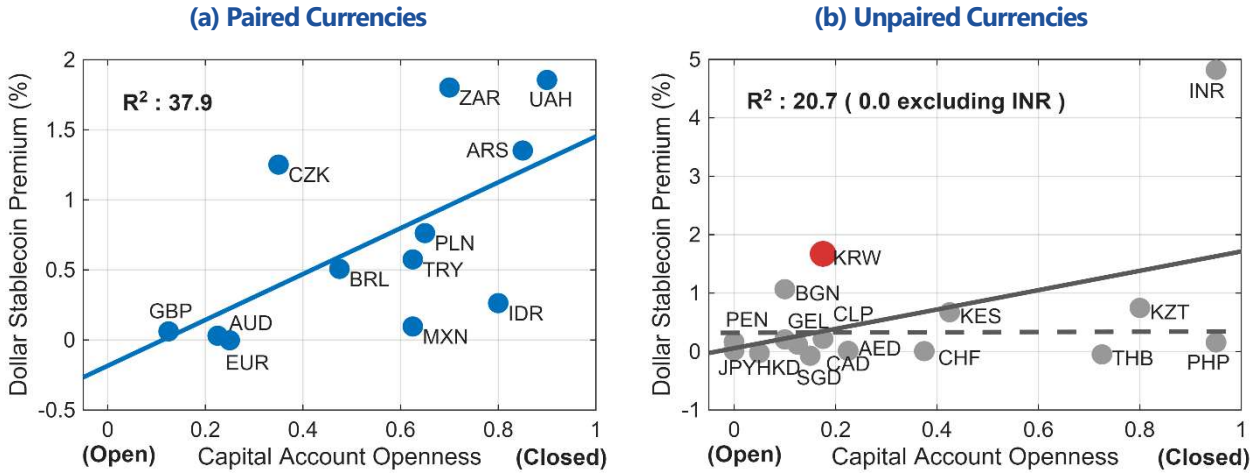
34. This cross-country heterogeneity cannot be fully explained by differences in capital account openness alone. Previous studies argue that countries with tighter capital controls have stronger incentives to circumvent these restrictions through less regulated cryptoasset markets, and that currency-specific cryptoasset premia reflect such demand (Makarov and Schoar, 2020; Graf von Luckner, Koepke, and Sgherri, 2024). Korea, however, has an active cryptoasset market and a relatively open capital market, yet exhibits a high USD stablecoin premium. This suggests that, in addition to capital account openness, the market structure linking trading across markets may be an important determinant of cross-country differences in USD stablecoin premia.

35. [Figure 10] plots the relationship between USD stablecoin premia and capital account openness by whether the currency is paired on Binance and reveals a clear difference between paired and unpaired currencies. Among paired currencies, lower capital account openness is strongly associated with higher USD stablecoin premia (R^2 = 37.9%)[Figure 10(a)]. Among unpaired currencies, however, this relationship is much weaker and, after excluding India as an outlier, has virtually no explanatory power (R^2 = 0.0%)[Figure 10(b)].

36. This difference suggests that the effect of capital account openness on USD stablecoin premia may depend on market structure. When a price gap arises between the FX and USD stablecoin markets, arbitrageurs with access to both markets can trade against it. Capital controls increase the transaction costs faced by these arbitrageurs. Stronger controls therefore raise arbitrage costs and allow the price gap, or USD stablecoin premium, to remain larger in equilibrium. Accordingly, when a trading structure links global FX and USD stablecoin markets, capital controls can be reflected in stablecoin premia through this arbitrage-cost channel. By contrast, when such linkages are limited, participation by arbitrageurs with access to both markets is also restricted, making price gaps more difficult to eliminate. In such markets, premia are more strongly influenced

by country-specific local supply and demand conditions than by capital controls, weakening the relationship between capital controls and USD stablecoin premia.

[Figure 10] USD Stablecoin Premia and Capital Account Openness by Currency



Note: The USD stablecoin premium is calculated as $(P_{LC}^{USDT} - S_{LC}/S_{LC})$, where P_{LC}^{USDT} is the price of USDT in the local currency and S_{LC} is the exchange rate against the U.S. dollar. The median premium over the period since 2022 is used. Capital account openness is measured using the most recent available value for each country; Germany's value is applied to the euro area.

Data: CryptoCompare, Bergant et al. (2026)

37. These findings suggest that a market structure connected to the global USD stablecoin market may be an important determinant of price integration between USD stablecoin and FX markets. The next section therefore uses the introduction of Binance fiat–stablecoin pairing to empirically examine the effects of this change in market structure.

Ⅳ. Empirical Results on Fiat Pair Listings

Ⅳ-1. Strengthening of Price Integration Following Fiat-Stablecoin Pairing

38. This section examines whether Binance fiat–stablecoin pairing strengthened price integration. Under the mechanism described above, local demand for USD stablecoins can be met more elastically through global intermediaries after pairing. This should facilitate arbitrage between the FX and USD stablecoin markets and reduce USD stablecoin premia on local exchanges.

39. To test this hypothesis, we estimate the following regression model.

$$p_{ijt} = \beta \mathbb{1}(\text{Paired})_{it} + \gamma' X_{it} + \alpha_{ij} + \delta_t + \varepsilon_{ijt}$$

p_{ijt} denotes the USD stablecoin premium for fiat currency i. It is measured as the log deviation between the price of USD stablecoin j, denominated in local currency i, and the nominal exchange rate, aggregated to the weekly frequency using exchange-level stablecoin trading volume as weights. $\mathbb{1}(Paired)_{it}$ is an indicator equal to one for periods after Binance first introduced trading between fiat currency i and a USD stablecoin, either USDT or USDC. α_{ij} denotes currency-by-stablecoin fixed effects, δ_t denotes week fixed effects, and X_{it} denotes country-specific market shock variables.

[Table 4] Changes in USD Stablecoin Premium Following Binance Fiat-Stablecoin Pairing

	Stablecoin Premium (%)			
	(1)	(2)	(3)	(4)
$\mathbb{1}(\text{Paired})_{it}$	-0.374***	-0.380***	-0.382***	-0.329***
	(0.075)	(0.077)	(0.077)	(0.081)
Observations	11,926	11,926	11,926	11,926
Within R^2	0.040	0.040	0.040	0.040
Currency FE	✓			
Pair FE		✓	✓	✓
Time(week) FE	✓	✓		
Time(month) FE			✓	
Stablecoin × Time(month) FE				✓
Global Controls			✓	✓

Notes: The dependent variable is the volume-weighted average USD stablecoin premium, p_{ijt}, defined as the log deviation, expressed in percent, between the price of USD stablecoin $j \in USDT, USDC$ denominated in currency i and the corresponding nominal exchange rate. We first calculate the premium for each exchange excluding Binance and then take a weighted average using exchange-level stablecoin trading volume. A higher value indicates that the USD stablecoin trades at a higher price relative to the spot exchange rate in that currency. $\mathbb{1}(\text{Paired})_{it}$ is an indicator equal to one for periods after Binance first introduced trading between the fiat currency and a USD stablecoin, either USDT or USDC. Pair FE denotes currency-by-stablecoin fixed effects. The control variables include indicators for the exchange rate adjustment by Argentina's Milei administration in December 2023, the Russia–Ukraine war period, and the declaration of martial law in Korea in December 2024, thereby controlling for country-specific market shocks unrelated to Binance pairing. In specifications without time fixed effects, log changes in the U.S. dollar index (DXY) and the VIX are included to control for global financial conditions. Standard errors are clustered at the currency-by-week level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

40. The results show that USD stablecoin premia declined significantly after Binance listed fiat–stablecoin trading pairs[Table 4]. Across all four specifications with different fixed-effect structures, the coefficient on the pairing indicator, $\mathbb{1}(\text{Paired})_{it}$ is significantly negative, ranging from −0.33 to −0.38 percentage points. This indicates that USD stablecoin premia on local exchanges declined significantly after Binance introduced fiat–stablecoin pairing. The findings suggest that pairing connected local exchanges to global intermediaries with access to both the FX market and global USD stablecoin liquidity, facilitating cross-market arbitrage and thereby strengthening price integration between USD stablecoin and FX markets.[12]

41. Under the mechanism described above, a larger premium differential between Binance and local exchanges—where the premium is higher on local exchanges—is expected to increase net inflows of USD stablecoins from Binance into the corresponding fiat–stablecoin market. This reflects arbitrage based on the price differential between Binance and local exchanges, which requires transfers across the two markets. By contrast, transfers between exchanges that do not involve Binance are not directly part of this arbitrage channel and are therefore not expected to exhibit a systematic relationship with the Binance–local exchange premium differential.

42. To test this prediction, we estimate the relationship between the USD stablecoin premium differential between local exchanges and Binance and two dependent variables: net inflows from Binance and net inflows from exchanges other than Binance, as defined in the previous section. The analysis covers the 12 fiat currencies for which Binance supports fiat–stablecoin trading. This restriction is necessary because calculating the premium differential between local exchanges and Binance requires Binance trading prices for the same fiat currency. The regression model is specified as follows.

$$NetInflow_{ijt}^{Binance} = \beta\tilde{\Delta}_{ijt}(Spread) + \gamma' X_{it} + \alpha_{ij} + \delta_t + \varepsilon_{ijt}$$

$$Net\ Inflow_{ijt}^{\neq Binance} = \beta\tilde{\Delta}_{ijt}(Spread) + \gamma' X_{it} + \alpha_{ij} + \delta_t + \varepsilon_{ijt}$$

$Net\ Infow_{ijt}$ denotes net inflows of stablecoin j from Binance or other exchanges into currency market i.[13] $\tilde{\Delta}_{ijt}(Spread)$ is the difference between the exchange-volume-weighted average USD stablecoin premium for fiat currency i and the corresponding premium on Binance, standardized by the time-series standard deviation for each currency-stablecoin pair (i, j). α_{ij} denotes currency-by-stablecoin fixed effects, δ_t denotes week fixed effects, and X_{it} denotes country-specific market shock variables.

43. The results show that a larger USD stablecoin premium on local exchanges relative to Binance is associated with significantly greater net inflows of USD stablecoins from Binance

[12] As a robustness check, event-study estimates show that the coefficients over the 18 weeks before pairing are jointly insignificant (F-test p-value = 0.8), providing no evidence of differential pre-trends between paired and unpaired currencies. In addition, when the sample is divided into early adopters (2019–2020) and recent adopters (2023–2025), the difference in estimated effects between the two groups is not statistically significant (p = 0.2). These results suggest that the findings are not driven by a particular adoption period or a small subset of countries.

[13] The net stablecoin inflow variable from exchanges other than Binance is constructed using the method described in [Appendix 2].

into the corresponding fiat currency market[Table 5]**.** By contrast, net inflows from other exchanges are not significantly related to the premium differential. This suggests that price gaps within a given fiat currency market are primarily adjusted through USD stablecoin transfers involving Binance. The findings are therefore consistent with the mechanism through which Binance fiat–stablecoin pairing strengthens price integration.

[Table 5] Local-Binance Premium Differentials and USD Stablecoin Flows Across Exchanges

	Net Inflow from Source X divided by Total USD Stablecoin Transfers		*Net Inflow from Source X divided by Gross Flow of Currency i*	
Source X	Binance	Others	Binance	Others
	(1)	(2)	(3)	(4)
$\widetilde{\Delta}_{ijt}(Spread)$	0.0015***	-0.0003	0.589***	-0.362
	(0.0004)	(0.0005)	(0.228)	(0.252)
Observations	2,552	2,552	2,552	2,552
Within R^2	0.012	0.003	0.007	0.042
Pair FE	✓	✓	✓	✓
Time(week) FE	✓	✓	✓	✓

Notes: The dependent variables in columns (1) and (3), labeled Binance, are net inflows of USD stablecoin $j \in \{USDT, USDC\}$ from Binance into fiat currency market i, scaled either by the total inflows and outflows of stablecoin j or by the sum of inflows into and outflows from currency market i. The dependent variables in columns (2) and (4), labeled Others, are net inflows from exchanges other than Binance into local exchanges, scaled in the same manner. $\widetilde{\Delta}_{ijt}(Spread)$ is the difference between the exchange-volume-weighted average premium of USD stablecoin $j \in \{USDT, USDC\}$ in fiat currency i and the corresponding Binance premium, standardized by the standard deviation for each currency-stablecoin pair (i, j). A positive value indicates that USD stablecoin prices are higher on local exchanges than on Binance. Pair FE denotes currency-by-stablecoin fixed effects. The control variables include indicators for the exchange rate adjustment by Argentina's Milei administration in December 2023 and the Russia-Ukraine war period, thereby controlling for country-specific market shocks unrelated to Binance pairing. Standard errors are clustered at the currency-by-week level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

IV-2. Stronger Shock Transmission Following Fiat-Stablecoin Pairing

44. Under the mechanism described above, after Binance introduces fiat–stablecoin pairing, demand pressure in the USD stablecoin market is transmitted to the FX market as professional market makers manage the fiat currency positions generated by trading.

45. This demand pressure is difficult to observe directly. This is because the demand for and supply of USD stablecoins are jointly determined, making it difficult to distinguish demand shocks from supply shocks using observed trading volume alone.

46. We therefore use two proxies for demand pressure in the USD stablecoin market. The first is the USD stablecoin premium, a price-based measure that increases when demand for USD stablecoins exceeds supply. The second is net buyer-initiated order flow, a volume-based measure that reflects the fiat currency positions taken on by market makers as counterparties to trades.

47. We first use the USD stablecoin premium as a price-based measure and estimate the following regression model to examine whether its effect on exchange rates increases after Binance introduces fiat–stablecoin pairing.

$$\Delta \log S_{it} = \beta_1 p_{ijt-} + \beta_2 \left(p_{ijt-1} \times \mathbb{1}(\text{Paired})_{it}\right) + \beta_3 \mathbb{1}(\text{Paired})_{it} + \gamma' X_{it} + \alpha_{ij} + \delta_t + \varepsilon_{ijt}$$

where $\Delta \log S_{it}$ denotes the log change in the nominal exchange rate of fiat currency i against the U.S. dollar. p_{ijt-1} denotes the lagged premium of USD stablecoin j in fiat currency i, calculated as an exchange-volume-weighted average. $\mathbb{1}(\text{Paired})_{it}$ is an indicator equal to one after Binance introduces trading between the fiat currency and a USD stablecoin. $p_{ijt-1} \times \mathbb{1}(\text{Paired})_{it}$ is the interaction between the lagged premium and the post-pairing indicator. α_{ij} denotes currency-by-stablecoin fixed effects, δ_t denotes week fixed effects, and X_{it} denotes country-specific market shock variables.

48. The estimates[Table 6] are consistent with the mechanism described above: Binance fiat–stablecoin pairing strengthens the transmission of demand pressure from the USD stablecoin market to the FX market. Before pairing, the relationship between USD stablecoin premia and exchange rate changes is not statistically significant. After pairing, however, this relationship becomes significantly stronger. Specifically, the coefficient on the interaction between the premium and the pairing indicator is positive in all specifications and statistically significant, ranging from approximately 0.13 to 0.15. This implies that the same increase in the USD stablecoin premium leads to a larger increase in the exchange rate—that is, greater local currency depreciation—after pairing.

[Table 6] Strengthening of Shock Transmission Following Binance Fiat-Stablecoin Pairing

	$\Delta \log S_{it}$ (%)			
	(1)	(2)	(3)	(4)
p_{ijt-1}	0.016	0.015	0.016	0.018
	(0.012)	(0.012)	(0.012)	(0.012)
$p_{ijt-1} \times \mathbb{1}(\text{Paired})_{it}$	0.129**	0.133**	0.147***	0.149***
	(0.054)	(0.054)	(0.055)	(0.055)
$\mathbb{1}(\text{Paired})_{it}$	0.088	0.082	0.087	0.078
	(0.067)	(0.064)	(0.064)	(0.064)
Observations	11,842	11,842	11,842	11,842
Within R^2	0.014	0.015	0.116	0.116
Currency FE	✓			
Pair FE		✓	✓	✓
Time(week) FE	✓	✓		
Time(month) FE			✓	
Stablecoin × Month FE				✓
Market Controls			✓	✓

Note: The dependent variable, $\Delta \log S_{it}$, is the weekly log change in the nominal exchange rate of fiat currency i against the U.S. dollar (%), where a positive value indicates depreciation of the local currency. p_{ijt-1} denotes the one-period-lagged premium of USD stablecoin $j \in \{USDT, USDC\}$ in fiat currency i, calculated as an exchange-volume-weighted average. $p_{ijt-1} \times \mathbb{1}(\text{Paired})_{it}$ is the interaction between the lagged premium and an indicator equal to one after Binance introduces trading between the fiat currency and a USD stablecoin, either USDT or USDC. The coefficient on the interaction term measures the change in the effect of the USD stablecoin premium on the exchange rate after pairing. Pair FE denotes currency-by-stablecoin fixed effects. The control variables include indicators for the exchange rate adjustment by Argentina's Milei administration in December 2023 and the Russia–Ukraine war period, thereby controlling for country-specific market shocks unrelated to Binance pairing. Standard errors are clustered at the currency-by-week level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

49. We next use net buyer-initiated order flow for Binance-paired currencies and Korean exchanges that support USDT trading to examine its effects on USD stablecoin premia and exchange rates. Under the mechanism described above, the same stablecoin demand shock is expected to be reflected mainly in a higher premium in markets without pairing. In paired markets, by contrast, it is more likely to be transmitted to the FX market through professional market makers' position management and reflected in exchange rate movements. This linkage is absent in Korea because Binance does not support KRW–stablecoin pairing, but it can operate for currencies paired on Binance. Comparing the two therefore allows us to examine how the effects of order flow differ depending on the availability of Binance fiat–stablecoin pairing.[14]

50. To test this prediction, we estimate the following regression model separately for the panel of Binance-paired currencies and for Korea.

$$Y_{ijt} = \beta \cdot \textit{Net Buying Pressure}_{ijt} + \gamma' X_{it} + \alpha_{ij} + \delta_t + \varepsilon_{ijt}$$

The dependent variable Y_{ijt} is either the daily log change in the nominal exchange rate of fiat currency i against the U.S. dollar or the daily premium of USD stablecoin j denominated in fiat currency i. $Net\ Buying\ Pressure_{ijt}$ denotes daily net buying pressure in fiat-stablecoin trading. α_{ij} denotes currency-by-stablecoin fixed effects, δ_t denotes day fixed effects, and X_{it} denotes country-specific market shock variables, including the growth rate of trading volume.

[Table 7] Effects of USD Stablecoin Net Buying Pressure by Pairing Status

	$\Delta \log S_{it}$ (%)		USD Stablecoin Premium (%)	
Currency	Binance-Paired	KRW	Binance-Paired	KRW
Net Buying Pressure	0.037***	-0.006	0.086***	0.398***
	(0.006)	(0.017)	(0.013)	(0.068)
Partial R^2 (%)	0.21	0.02	0.26	4.96
Observations	19,059	661	19,059	661
Currency FE	✓		✓	
Stablecoin FE	✓		✓	
Time(date) FE	✓		✓	

Note: The Binance-paired currency sample includes the currencies of Australia, Brazil, Czechia, the United Kingdom, Indonesia, Mexico, Poland, Türkiye, Ukraine, and South Africa, as well as the euro. Exchange rate changes, $\Delta \log S_{it}$, are measured as log changes in nominal exchange rates against the U.S. dollar. The USD stablecoin premium is the premium of USD stablecoin $j \in \{USDT, USDC\}$ denominated in fiat currency i. For Binance-paired currencies, $Net\ Buying\ Pressure$ is calculated as buyer-initiated order flow minus seller-initiated order flow in the corresponding fiat–stablecoin market, divided by lagged trading volume. For Korea, it is calculated analogously using USDT trading on Korean exchanges. To facilitate cross-country comparisons, $Net\ Buying\ Pressure$ is standardized by the standard deviation for each currency–stablecoin pair (i, j). All variables are based on daily data. Standard errors are heteroskedasticity-robust and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

[14] Buyer-initiated order flow data are available only for currencies paired on Binance and Korean exchanges that support USDT trading. Because Binance does not support KRW–USD stablecoin pairing, Korea serves as a control group in which the mechanism examined in this study does not operate.

51. The estimates[Table 7] reveal clear differences in the effects of net buying pressure on exchange rates and USD stablecoin premia depending on the availability of Binance fiat–stablecoin pairing. For currencies paired on Binance, stronger net buying pressure significantly raises the exchange rate, indicating local currency depreciation, while its effect on the premium is significant but relatively small (0.086). In Korea, by contrast, net buying pressure has no significant effect on the exchange rate but substantially increases the USD stablecoin premium (0.398). The partial R^2 of net buying pressure is also higher for exchange rates among Binance-paired currencies and for USD stablecoin premia in Korea, reinforcing the same conclusion. These findings are consistent with the mechanism described above: the same stablecoin demand shock is transmitted to the FX market when fiat–stablecoin pairing is available, but is absorbed mainly within the local USD stablecoin market when it is not.[15]

[15] Country-level coefficient estimates are reported in [Appendix 3].

Ⅴ. Conclusion

52. This study exploits the introduction of Binance trading between specific fiat currencies and USD stablecoins, which creates a market structure in which global intermediaries participate as trading counterparties, to examine how this structural change affects the linkage between USD stablecoin and FX markets. The results show that the market structure of fiat–stablecoin trading is an important determinant of cross-market linkages. Even for otherwise identical trades, the degree of linkage with the FX market can differ depending on the composition of trading counterparties.

53. Specifically, we analyze the linkage between USD stablecoin and FX markets along two dimensions: price integration and shock transmission. Price integration refers to the extent to which price gaps between the two markets are reduced through arbitrage. Shock transmission refers to the extent to which demand shocks in the USD stablecoin market affect exchange rates through FX transactions.

54. The results show that the linkage between the two markets strengthens along both dimensions after Binance introduces fiat–stablecoin pairing. Following pairing, USD stablecoin premia on local exchanges decline, while price gaps between local exchanges and Binance are associated with greater stablecoin inflows from Binance, indicating stronger price integration through cross-exchange arbitrage. The relationship between stablecoin premia, which reflect demand pressure in the USD stablecoin market, and exchange rates also strengthens after pairing, indicating greater transmission of stablecoin demand shocks to the FX market. The analysis of net buyer-initiated order flow yields similar results. For Binance-paired currencies, stronger USD stablecoin demand leads to higher exchange rates, whereas in Korea, where pairing is unavailable, its effects are reflected mainly in higher USD stablecoin premia.

55. These findings show that when global intermediaries participate as counterparties in fiat–stablecoin trading, the linkage between USD stablecoin and FX markets can strengthen through both price integration and shock transmission.

56. The results suggest that future changes in the structure of Korea's cryptoasset market could affect how USD stablecoin and FX markets are linked. At present, participation by for-profit corporations and foreign investors in Korea's cryptoasset market is restricted, limiting the role of global intermediaries that connect the FX and USD stablecoin markets. As a result, the channels through which USD stablecoin trading is directly linked to the FX market remain limited. However, if future regulatory changes expand corporate and foreign participation, demand shocks in the USD stablecoin market may involve greater FX trading and have a more direct effect on exchange rates.

57. Accordingly, reforms to the digital asset regulatory framework, the internationalization of the Korean won, and improvements in the structure of the FX market should be approached as interrelated policy tasks. Broader participation in the cryptoasset market following regulatory changes could strengthen the linkage between USD stablecoin and FX markets. By contrast, won internationalization and FX market reforms could mitigate the resulting shocks by broadening the market participant base and increasing FX market liquidity. Future policy design should therefore consider both the potential for stronger linkages between cryptoasset and FX markets and the FX market's capacity to absorb shocks.

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[Appendix 1] Data Description

The data used in this study and their sources are summarized in [Table A1-1].

[Table A1-1] Data Source

Data	Source
Fiat–USD Stablecoin Pairing Dates and Net Buyer-Initiated Order Flow	Binance Data Collection
Fiat–USD Stablecoin Prices and Trading Volumes	CCXT, CoinAPI, CryptoCompare
Spot Exchange Rates	Bloomberg
USD Stablecoin Inflows and Outflows	Chainalysis
Capital Account Openness	Bergant et al. (2026)
Parallel Exchange Rate for Argentina	https://bluedollar.net/informal-rate/
Parallel Exchange Rate for Ukraine	https://minfin.com.ua/ua/currency/

The key variables are constructed as follows.

[Table A1-2] Key Variables

Variables	Construction
$\mathbb{1}\,(\text{Paired})_{it}$	An indicator equal to one if USDT or USDC can be traded against fiat currency i
$Spread_{it}$	The percentage deviation of the exchange-volume-weighted average USD stablecoin price denominated in fiat currency i from the spot exchange rate
$\widetilde{\Delta}_{ijt}(Spread)$	The difference between the spread for fiat currency i, calculated excluding Binance, and the corresponding spread on Binance
$Net\ Inflow$	Constructed by allocating Chainalysis data on daily USD stablecoin transfers across exchanges according to each exchange's shares of fiat currency trading volume
$Net\ Buying\ Pressure$	The difference between daily buyer-initiated and seller-initiated trading volume, scaled by the previous day's trading volume
$Capital\ Account\ Openness$	A measure provided by Bergant et al. (2026), who apply an LLM to the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER) to estimate capital account openness by country and year

[Appendix 2] Net Inflows from Exchanges Other Than Binance

Net inflows from all exchanges other than Binance (non-Binance exchanges) are estimated through the following procedure.

1) USD Stablecoin Inflows from a Specific Exchange into Fiat–Stablecoin Market i

We first define inflows of USD stablecoin $j \in \{USDT, USDC\}$ from origin exchange ko to destination exchange kd that are allocated to fiat-stablecoin market i, denoted by $Inflow_{ijt}^{ko \to kd}$ as follows,

$$Inflow_{ijt}^{ko \to kd} = Flow_{jt}^{ko \to kd} \omega_{ikd}$$

where $Flow_{jt}^{ko \to kd}$ denotes the volume of USD stablecoin j transferred from exchange ko to exchange kd, and ω_{ikd} denotes the share of currency i in trading volume on destination exchange kd. For an exchange that supports only one fiat currency, all USD stablecoins transferred into the exchange are assigned to that fiat–stablecoin market. If the destination exchange supports trading between USD stablecoins and multiple fiat currencies, the transferred amount is allocated across fiat–stablecoin markets according to each fiat currency's share of the exchange's fiat–stablecoin trading volume. This procedure converts blockchain-based data on stablecoin transfers across exchanges into fiat-currency-level inflow measures.

By summing these inflows, calculated for each origin exchange, destination exchange, and fiat–stablecoin market, across all destination exchanges, we estimate the total volume of USD stablecoins flowing from a given origin exchange into fiat–stablecoin market i.

$$Inflow_{ijt}^{ko} = \sum_{kd} Flow_{jt}^{ko \to kd} \omega_{ikd}$$

2) USD Stablecoin Outflows from Fiat–Stablecoin Market i to a Specific Exchange

Using the same approach, we calculate outflows from a specific fiat–stablecoin market to a given exchange. The portion of USD stablecoin j transferred from origin exchange ko to destination exchange kd that is attributed to outflows from fiat–stablecoin market i, denoted by $Outflow_{ijt}^{ko \to kd}$ is defined as follows,

$$Outflow_{ijt}^{ko \to kd} = Flow_{jt}^{ko \to kd} \omega_{iko}$$

where, $Flow_{jt}^{ko \to kd}$ denotes the volume of USD stablecoin j transferred from origin exchange ko to destination exchange kd, while ω_{iko} denotes the share of currency i in trading volume on origin exchange ko. As in the case of inflows, if an exchange supports only one fiat currency, all outflows from that exchange are assigned to the corresponding fiat–stablecoin market. If an exchange supports multiple fiat currencies, its outflows are allocated across fiat–stablecoin markets in proportion to the trading-volume share of each fiat currency. Using this procedure, the volume of outflows from fiat–stablecoin market i to destination exchange kd is estimated as follows.

$$Outflow_{ijt}^{kd} = \sum_{ko} Flow_{jt}^{ko \to kd} \omega_{iko}$$

3) Net Inflows from All Non-Binance Exchanges into Fiat–Stablecoin Market i

Net inflows are defined as USD stablecoin inflows from non-Binance exchanges into the market less USD stablecoin outflows from the market to non-Binance exchanges.

Inflows into fiat–stablecoin market i from exchanges other than Binance are based on USD stablecoin transfers for which both the origin and destination exchanges are non-Binance exchanges. Specifically, each inter-exchange transfer is allocated according to the destination exchange's fiat-currency shares of USD stablecoin trading volume and then summed across all non-Binance exchange pairs (ko, kd), as follows.

$$\text{Inflow}_{ijt}^{\neq binance} = \sum_{ko \neq binanc} \sum_{kd \neq binanc} Flow_{jt}^{ko \to kd} \omega_{ikd}$$

Outflows from fiat–stablecoin market i to exchanges other than Binance are calculated by allocating USD stablecoin transfers for which both the origin and destination exchanges are non-Binance exchanges according to the origin exchange's fiat-currency shares of USD stablecoin trading volume and then summing across all non-Binance exchange pairs (ko, kd), as follows.

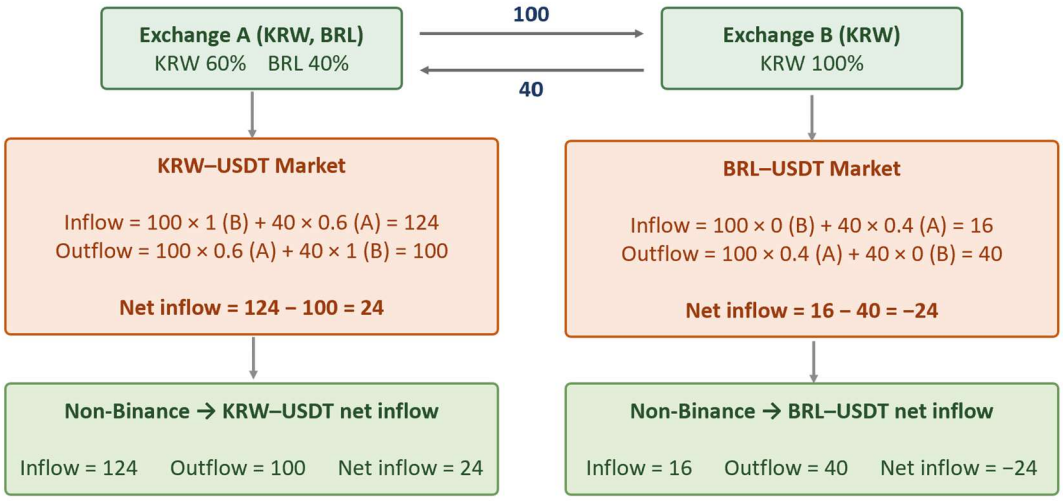
$$Outflow_{ijt}^{\neq binance} = \sum_{ko \neq binance} \sum_{kd \neq binance} Flow_{jt}^{ko \to kd} \omega_{iko}$$

Net inflows from all exchanges other than Binance into fiat–stablecoin market i are defined as the difference between inflows and outflows, as follows.

$$NetInflow_{ijt}^{\neq binanc} = \sum_{ko \neq binance} \sum_{kd \neq binance} Flow_{jt}^{ko \to kd} (\omega_{ikd} - \omega_{iko})$$

[Figure A2-1] illustrates the procedure used to calculate net inflows from non-Binance exchanges into a specific fiat–stablecoin market. USD stablecoins transferred across exchanges are allocated as inflows into each fiat–stablecoin market according to the destination exchange's fiat-currency trading shares. Conversely, they are allocated as outflows from each market according to the origin exchange's trading shares. Net inflows are then calculated by subtracting these outflows from the corresponding inflows.

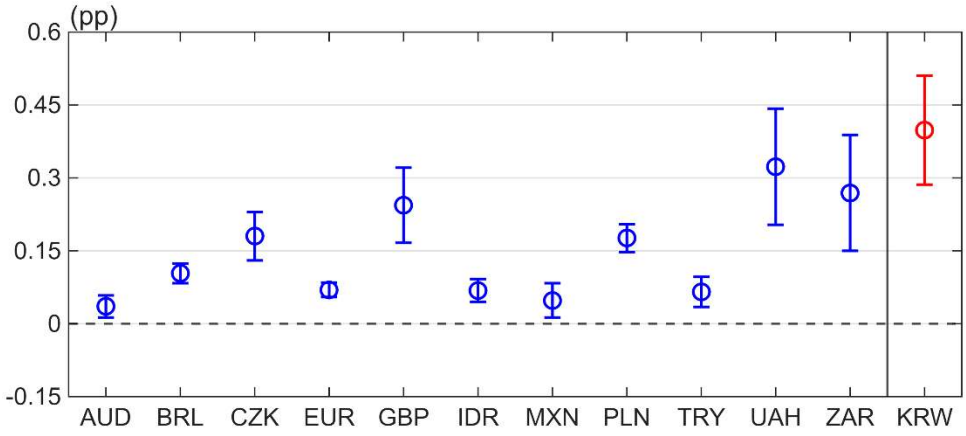
[Figure A2-1] Net Inflows from Exchanges Other Than Binance into Fiat–Stablecoin Market i



Note: The arrows indicate the direction of USD stablecoin transfers, and the numbers on the arrows indicate the transfer amounts.

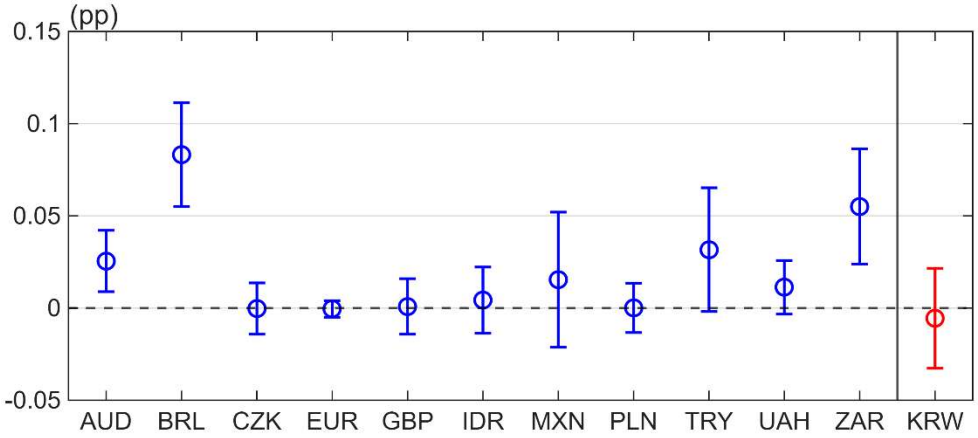
[Appendix 3] Currency-Level Heterogeneity in the Effects of USD Stablecoin Net Buying Pressure on Premia and Exchange Rates

[Figure A3-1] Cross-Currency Comparison: USD Stablecoin Net Buying Pressure → USD Stablecoin Premia



Note: The figure reports the estimated coefficients and 90% confidence intervals from regressions of daily USD stablecoin premia on currency-level net buying pressure. Binance-paired currencies comprise the Australian dollar (AUD), Brazilian real (BRL), Czech koruna (CZK), British pound (GBP), Indonesian rupiah (IDR), Mexican peso (MXN), Polish złoty (PLN), Turkish lira (TRY), Ukrainian hryvnia (UAH), South African rand (ZAR), and euro (EUR). For Binance-paired currencies, net buying pressure is calculated as buyer-initiated order flow less seller-initiated order flow in the corresponding fiat–stablecoin market, scaled by lagged trading volume. For the Korean won (KRW), it is calculated as buyer-initiated order flow less seller-initiated order flow on domestic exchanges supporting USDT trading, scaled by lagged trading volume. To facilitate cross-currency comparison, net buying pressure for each currency is standardized by its currency-specific standard deviation.
Data: Binance, Upbit

[Figure A3-2] Cross-Currency Comparison: USD Stablecoin Net Buying Pressure → Exchange Rates



Note: The figure reports the estimated coefficients and 90% confidence intervals from regressions of daily changes in the nominal bilateral exchange rate against the U.S. dollar, measured as local currency per U.S. dollar, on currency-level net buying pressure. Binance-paired currencies comprise the Australian dollar (AUD), Brazilian real (BRL), Czech koruna (CZK), British pound (GBP), Indonesian rupiah (IDR), Mexican peso (MXN), Polish złoty (PLN), Turkish lira (TRY), Ukrainian hryvnia (UAH), South African rand (ZAR), and euro (EUR). For Binance-paired currencies, net buying pressure is calculated as buyer-initiated order flow less seller-initiated order flow in the corresponding fiat–stablecoin market, scaled by lagged trading volume. For the Korean won (KRW), net buying pressure is calculated in the same manner using domestic exchanges supporting USDT trading. To facilitate cross-currency comparison, net buying pressure for each currency is standardized by its currency-specific standard deviation.
Data: Binance, Upbit, Bloomberg